Tarun R

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Research

My research lies at the intersection of 3D computer vision and robotics, with a focus on robot perception algorithms that enable reliable autonomous navigation. Currently, I am particularly interested in visual-inertial SLAM and relocalization.

Education

2025 – Present: MS by Research in Computer Science
International Institute of Information Technology (IIIT) Hyderabad, India
Advisor: Prof. Madhava Krishna
GPA: 9.50/10.00
Thesis: Robot Scene Understanding and Relocalization (in progress)

2020 – 2024: B.Tech in Mechatronics Engineering
SRM Institute of Science and Technology, Kattankulathur, India
GPA: 8.24/10.00
Thesis: Campus-Scale Autonomous Ackermann-Steering Robot

Publications and Patents

SILICA: Repurposing Diffusion Priors for Joint Glass Segmentation and Depth Estimation Project Page | Code | arXiv
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
***Tarun R**, Anuj Verma, Laksh Nanwani, Sourav Garg, Madhava Krishna*

CrowdSurfer: Sampling Optimization Augmented with Vector-Quantized Variational AutoEncoder for Dense Crowd Navigation Project Page | Code | arXiv
IEEE International Conference on Robotics and Automation (ICRA), 2025
Naman Kumar, Antareep Singha*, Laksh Nanwani*, Dhruv Potdar, **Tarun R**, Fatemeh Rastgar, Simon Idoko, Arun Kumar Singh, Madhava Krishna*

System and Method for 3D Point Cloud Rectification Using Depth Sensor with Revolved Reflective Optics
Indian Patent Application No. 202541131469 (filed, 2025)
*Laksh Nanwani, **Tarun R**, Madhava Krishna*

Teaching and Academic Service

Aug 2026 – Present: Teaching Assistant, Mobile Robotics (CS7.503), taught by Prof. Madhava Krishna
IIIT Hyderabad
May 2026 – Jun 2026: Guest Lecturer, Visual and LiDAR SLAM
Robotics Research Center (RRC) Summer School 2026
May 2025 – Jun 2025: Guest Lecturer, Robot Operating System (ROS 2) and Capstone Project
Robotics Research Center (RRC) Summer School 2025
Reviewing: Co-reviewer for IEEE Robotics and Automation Letters (RA-L) 2025 and IEEE International Conference on Robotics and Automation (ICRA) 2027

Projects

2026: Traversability-Aware Indoor Navigation Slides
- Used diffusion foundation model priors for indoor floor segmentation and depth estimation.
- Wrote a Numba-based GPU MPPI controller, soft-constrained to generate trajectories only within the traversable region of the segmentation.

2026: Transparent Object Grasping with a Modified SILICA Slides
- Trained SILICA with CLIP conditioning to predict glass objects and glass scenes separately, resulting in far fewer false positives.
- Used the model's transparent-object segmentation and depth to generate grasps with NVIDIA GraspGen, and wrote an RRT* planner to reach and pick up the object.

2025: Omnidirectional Vision-LiDAR Calibration and Fusion Framework Slides

- Wrote a C++ ROS 2 driver and image stitcher for the Insta360 camera. Since the camera does not timestamp its images, designed a synchronization method from scratch to obtain output that is synchronized and calibrated with a Livox MID-360 LiDAR.
- Built a targetless extrinsic calibration pipeline for a LiDAR, five RealSense cameras, and an Insta360 camera using the LightGlue matcher.
- Used this setup to collect over 2 TB of multi-sensor data in a shopping mall.

2025: Surfing Scout the Quadruped Slides

- Designed a 12-servo controller PCB and wrote firmware in C with micro-ROS to interface with the higher-level stack.
- Wrote a ROS 2 driver for the 12 actuators and implemented inverse kinematics from the body to the leg tips, enabling the quadruped to reach target body orientations.
- Wrote a C++ controller that uses a single IMU to accurately track and follow target body motions, and explored modeling the walking gait using Hilbert's swing.

Experience

Jul 2025 – Present: Hardware Lab Administrator

Robotics Research Center (RRC), IIIT Hyderabad

- Manage the robotics hardware lab, which serves 40+ students, including maintaining the lab's 3D printers and modernizing the robots' power systems and PCBs.
- Maintain CI/CD pipelines for multiple ground robots, and migrated them from legacy ROS versions to current releases.

Jun 2024 – Jul 2025: Research Fellow Videos | Code

Robotics Research Center (RRC), IIIT Hyderabad

- Built Phase 2 of the Autonomous Wheelchair Project from the ground up, making all system design choices and developing the full 3D LiDAR-based hardware and software stack on an NVIDIA Jetson Orin NX.
- Developed an MPPI controller and a 3D LiDAR-based mapping and relocalization stack, and dockerized the setup with CI/CD pipelines so that it could be ported to and deployed on multiple wheelchairs.
- Designed the sensor and battery mounting hardware, as well as a power distribution PCB.
- Deployed the system in shopping malls, the GMR Airport Innovex center at Hyderabad, office buildings, academic blocks, and other public spaces.

Jan 2025 – Oct 2025: Systems and Mechanical Engineer Videos | Code

The Energy Labs, India

- Designed several in-house quadrotors from scratch for different payloads and weights, and manufactured them for sale as commercial products.
- Developed ROS 2 system integration for GPS-denied flight using visual-inertial odometry.

Nov 2020 – Jul 2022: Drivetrain Engineer

Team 1.618, Formula Student Team, SRM Institute of Science and Technology

- Led drivetrain engineering for a 30-member student team, designing and fabricating a hybrid Formula Student car for national and international competitions.
- Designed and fabricated a planetary coupling system, optimized an eccentric chain tensioner, and managed sponsor relationships.

Awards and Certifications

Nov 2021: Overall Champions, Formula Imperial (7th season), organized by the Imperial Society of Innovative Engineers (ISIE India), with Team 1.618
Credential ID: ISIE/VFI2021/W/076

Aug 2021: Certified SolidWorks Associate (CSWA), Mechanical Design, Dassault Systèmes
Credential ID: C-CNMJ6H3WEP

Skills

Technical: Python, C/C++, Linux, ROS/ROS 2, PyTorch, Numba, JAX, CasADi, OpenCV, Rerun, SolidWorks
Languages: English (fluent), Tamil (native), Telugu, Hindi